

Where the Seconds Go: Bounding Mixture-of-Experts Decode from Host Memory, and Measuring What Foresight Is Worth

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Abstract—A Mixture-of-Experts model too large for one consumer GPU keeps most experts in host memory, so decoding one token at a time waits on host reads. We bound how fast such decoding can be on a given machine, measure where the time goes, and price foresight of the routing. The bound is the fewest host reads any expert cache of a given size can make (Belady’s MIN with bypass) at the machine’s probed read rate. Our llama.cpp expert cache, faster than two other systems at most settings, reaches 31–54% of the bound’s speed on consumer machines at the smallest cache we test. Oracles in the engine then change what is cached (the experts MIN keeps, not ours), how each admitted expert is read (once, copied into its GPU slot, instead of twice, by the CPU and a background copy), and when. A per-machine table that also copies misses into slots the cache did not choose hides the first change: with it, MIN’s set pays only when read once; without it, on five machines, MIN’s set alone closes 19% of the gap and both changes together 36%. Even reading ahead leaves about half of the gap. On five fast-link machines, four policies without foresight built from the engine’s own parts recover at most 6% of an oracle’s gain. In trace replay, half of foresight’s value needs the routing of as many distinct experts as 0.65 of a full cache. Registered predictions were committed before their machines were rented; the consumer range and the replay result are post hoc.

Index Terms—Caching, foresight, mixture of experts, offloading, performance bounds

I. INTRODUCTION

Open-weight Mixture-of-Experts (MoE) models now carry 30–120 B parameters and activate a few billion per token; a desktop GPU has 24–32 GB. Running such a model for one user therefore means keeping most experts in host DRAM. Each decoded token reads its routed experts from GPU memory if they are resident, and otherwise from host memory: either the CPU runs the expert, or the expert is copied over PCIe into a GPU slot [1], [2].

Many systems build on this: GPU expert caches [3]–[5], CPU execution of misses [6]–[8], a per-miss choice between the two [9], [10] and expert prediction [11]–[13]. Each reports a speed-up over a baseline of its choosing. None says how fast the same machine could decode the model, or which part of the remaining time a better design could remove. Without that, a builder cannot tell whether to work on what is cached, how it is read, when it is read, or how far ahead the routing must be predicted.

Appendices A–N and the tables, figures and equations numbered S1, S2, ... are in the supplementary material available online.

We answer these questions for batch-1 decode of gpt-oss-120b [14] and Qwen3-30B-A3B [15] on rented RTX 5090 machines. We derive a bound, build oracles into a real engine that remove one part of the gap at a time, and run them on many machines, with every prediction committed before its machine started. Table I lists the answers and the evidence behind each.

Three terms carry the results. *MIN*, Belady’s optimal cache with bypass, decides which experts to keep, and the experts it keeps are *MIN*’s *set*. A missed expert is read from host memory *twice* when the CPU runs it and a background copy then fills its GPU slot, or *once* when it is copied into its slot in the step that needs it. And our engine’s deployed read path has *in-step fetches*: a per-machine table copies some misses into slots in the step, whether or not the cache chose them.

The headroom is large (Section III). *MIN*’s host reads at the machine’s probed rate bound the time per token of any system with C slots per layer, and that read time is nearly reachable in a microbenchmark. Our cache, which leads FreeToken and llama.cpp at most configurations, reaches 31–54% of the bound’s speed at the smallest gpt-oss budget on consumer machines (a scope set post hoc). Scored the same way on their papers’ hardware at datasheet rates (post hoc), ours reaches a median 27% and published systems 13.6%.

What is cached pays if the read path keeps it (Section IV). With the in-step fetches on, *MIN*’s set gains nothing unless each admitted expert is also read once instead of twice; then the two changes close 39% of the gap on fast-link machines (a post hoc subset) at the smallest gpt-oss budget. With them off, on five machines of two registered tests, *MIN*’s set alone closes 19% of the gap, reading the deployed set once 8%, and both together 36%: the set pays by itself; reading it once added most on the two machines with the fastest link for their CPU (post hoc).

Without foresight, little of it is reachable (Section VI). Four variants built from the engine’s admission margin and layer-ahead prediction recover at most 6% of the read-ahead oracle’s gain on five fast-link machines registered for the test: they read as many experts as the deployed cache.

The machine decides how foresight should be spent (Sections V and VII). Admit as few experts as optimality allows, keep them, and read them on the path the link-to-CPU ratio favours; where the link is fast, at the smallest budget,

TABLE I
OUR CLAIMS: THE THRESHOLD REGISTERED FOR EACH BEFORE ITS MACHINES STARTED, THE RESULT, AND A ONE-WORD VERDICT.

Claim	Registered threshold	Result	Verdict	Mach.	§
MIN’s reads at the probed rate bound any exact-routing cache with C slots per layer	derived, relative to the probe	no measured time on 25 launches is below it (fastest $1.33\times$ it); by the closed-form account (Appendix H), the engine’s reads imply a rate $> 5\%$ above the probe’s on 12 of 84 launch-budgets	derived	–	III
The bound’s read time is nearly reachable Our cache is faster than FreeToken and llama.cpp at most configurations	$\geq 50\%$ of it per layer (loose) on B (loose: after a pilot there): faster than FreeToken (interval > 1), $\geq 1.8\times$ llama.cpp ($\geq 2\times$ at Qwen3 12.5%); on S: within ± 0.06 of B’s ratios	86–96%, held (point); 5 of the job’s 33 clauses failed held on B; on S, five of six ratios fell outside ± 0.06 of B’s; faster at 11 of 12 in all	held (point) mixed	3 B, S	III III
With the in-step fetches off, MIN’s set pays alone	per machine, CPU-only path, by test: MIN-2R ≥ 1.05 (loose), $1.07\times$ at 11%, ≥ 1.15 , $1.20\times$ at 25%; Dep-1R 0.95, 0.98 to $1.10\times$; interaction > 0 ; pooled share $\geq 10\%$	MIN-2R $1.09\text{--}1.16\times$ and $1.24\text{--}1.36\times$, share 19% (held, mostly in-sample); Dep-1R $1.02\text{--}1.07\times$; interaction > 0 on four of five (failed on the Core i9-14900K); two checks failed where the fetch table was all zero	mixed	5	IV
Without the fetches, the fewest-admission set pays read twice	per machine: Few-2R $\geq 1.10\times$, MIN-2R $\geq 1.03\times$ a deployed cache without them; one read ≥ 0.05 more; deployed $0.80\text{--}1.00\times$	Few-2R $1.23\text{--}1.26\times$, MIN-2R $1.09\text{--}1.12\times$ (held); one read $+0.02\text{--}0.16$, deployed $0.98\text{--}1.07\times$ (each failed on the Core Ultra 9 285K, at 11%)	mixed	2	D
On the deployed read path, MIN’s set pays only together with one read (gpt-oss 11%, fast links, greedy schedule) Control: issuing the two-read arm’s copies a step earlier, to separate when a copy lands from the second read MIN’s fewest-admission set read once gains on stable machines	interaction > 0 on every machine; MIN-1R’s mean share of the gap $0.25\text{--}0.55$ (loose) misses $\leq 0.95\times$ Few-2R’s; slowdown ≤ 0.01 ; per machine and budget > 1 on every machine	positive on 9 of the panel’s 10 main-job machines (failed), 5 of 5 new ones (held; fast-link scope set in between); share 35% at 11%: misses $0.97\text{--}0.98\times$, $0.03\text{--}0.05$ slower; failed on 5 and 7 of 8 machine-budgets; so the control is uninformative	mixed failed	10 + 5 4	IV D
Policies without foresight recover little of the oracle’s gain The link-to-CPU ratio predicts the oracles’ speed on a second card Our cache reaches 31–54% of the bound on consumer machines (gpt-oss 11%) Published systems reach a median 13.6% of their bound Half of foresight’s value needs $\approx 0.65 C$ distinct experts ahead	speed $\leq 1.10\times$; capture $\leq 35\%$ (loose), per machine within 0.10 in log of the RTX 5090 trend (trend fitted post hoc) post hoc; consumer scope set after the data post hoc post hoc	best $1.007\text{--}1.024\times$, capture at most 6% (held) within 7% at ratios $0.37\text{--}0.77$ (held); new fast-link RTX 5090s off by up to 18% (post hoc) 31–54% on 25 consumer machines; 12–46% on server ones their papers’ hardware at datasheet rates on 9 models’ traces	held held post hoc post hoc post hoc	5 5 25 – –	VI VII III III VIII

Verdict: *held*, the 95% interval meets the threshold on every machine (*point*: only the point estimate does, or there is no interval); *mixed*, held in part and failed in part; *failed*; *derived*, follows from the definitions; *post hoc*, found in the data it describes, not registered. *Loose*: a threshold the earlier data made easy to meet. *Mach.*: machines that passed every check (for the row on the deployed read path, the panel’s main job and the new machines). Table S3 points to each claim’s data and clauses.

reading each admission once pays most. A trend in that ratio, fitted on RTX 5090s, predicted the oracles’ speed on five RTX 4090s within 7%; applied after the fact to the five new RTX 5090s, it missed by up to 18%.

Foresight has a price (Section VIII). In replay of 9 models’ routing traces (post hoc), half of its value needs the routing of about $0.65 C$ distinct experts per layer ahead; exact routing of the next token recovers at most a third of it on our workload.

Section IX turns these results into rules for system builders; Appendices D and N give every prediction, including the failures, and the artifact.

II. SETTING

Models and workload: gpt-oss-120b has 36 MoE layers of 128 experts, top-4, in MXFP4 (13.25 MB per expert). Qwen3-30B-A3B has 48 layers of 128 experts, top-8, in BF16 (9.44 MB per expert). Prompts are AIME-25 problems, and the cache carries over between problems, as in one user’s session. The system comparison decodes 256 tokens of each of the 30 problems through each system’s server. The engine experiments replay the model’s own greedy output, teacher-forced, so that every configuration sees the same routing. A

budget is the share C/E of each layer’s experts that fit in GPU slots: gpt-oss 11% and 25% are $C=14$ and $C=32$ of 128. The two smallest budgets of each model are *host-bound*: host reads, not the GPU, limit their speed (Section III gives the exceptions). Most results are at gpt-oss 11% and 25%.

The engine: Our cache is a 3,500-line patch to llama.cpp [16]. Each MoE layer has C GPU slots. A *miss* is read from host memory in one of two ways: CPU helper threads run it, or it is copied into a slot over PCIe and run on the GPU in the same step. A per-machine *fetch table*, computed from the machine’s bandwidth probe, sets how many of a layer’s misses are copied that way, whichever they are (the *in-step fetches*). The *deployed* policy admits a miss into a slot when its decayed request count beats the lowest resident’s by a margin $\kappa=1$. The CPU serves the admitted miss and a background copy then fills its slot, so the expert is read from host memory twice (Fig. 1); the copy is published at the start of the second step after the one that admitted it.

Terms: *MIN* is Belady’s optimum with bypass [17], [18]: it *admits* a miss only if some resident is needed later than the miss, which gives the most hits possible with C slots [19], [20]. Many schedules reach that optimum. The *greedy* one

TABLE II
THE NAMES THIS PAPER USES: CONFIGURATIONS (LEFT), MACHINE SETS AND YARDSTICKS (RIGHT).

Admitted set	read twice	read once	read ahead	<i>Machine sets</i> (RTX 5090 unless noted)	
<i>No foresight</i>				panel	the 15 machines that ran every state of Table IV before the registered tests; 10 of them ran its main job
deployed: decayed count beats the lowest resident's by $\kappa=1$ the same, larger margin ($\kappa=3$ at 11%, 2 at 25%)	deployed cache	Dep-1R	–	new machines	5 rented for the first registered test of Section IV (ratio ≥ 0.5)
the same, plus a layer-ahead copy ^a	Margin	–	–	fetches-off machines	5 rented for two later tests of Section IV (Table V)
every miss	LA	LA-1R, LA-1R-margin ^b	–	O3–O5	three machines that ran every budget (Section V)
<i>Oracles</i> (the future routing, from a trace recorded on the same machine)	–	admit every miss	–	hosts B, S	the system comparison (Table III)
MIN with bypass, greedy schedule	MIN-2R	MIN-1R	read-ahead oracle ^c	RTX 4090s	5 machines of the second-card tests (Section VII)
MIN with bypass, fewest admissions	Few-2R ^d	Few-1R	–	<i>Yardsticks</i> (T : time per token on one machine)	
Belady's rule, no bypass	Belady-2R	–	Belady-1R	speed	T_{deployed}/T ($\times$), the default
				share of the bound	T_{bound}/T , the bound being Eq. (1)
				share of the gap capture	$(T_{\text{deployed}} - T)/(T_{\text{deployed}} - T_{\text{bound}})$
				value of foresight	speed gain over the read-ahead oracle's gain share of MIN's host-read saving over admitting every miss

Read twice: the CPU serves the miss, then a background copy fills the slot; *read once*: fetched into the slot in the step. ^a A copy, one layer ahead, of the non-resident expert that the next layer's router, applied to this layer's input, ranks highest; an admission it predicted is read once. ^b LA-1R with Margin's margin. ^c MIN's admissions copied three steps ahead where the next use allows. ^d Few-2R-early: each copy issued one step earlier, so that it lands for the next use. *CPU only*: a configuration with the in-step fetches off, every miss served by the CPU (Section IV).

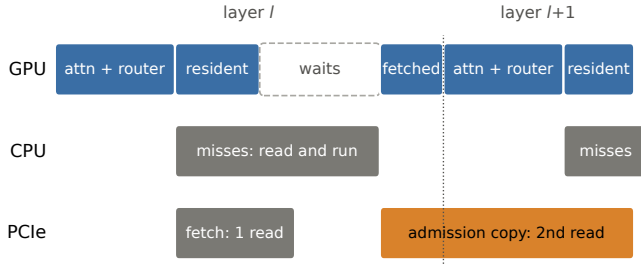


Fig. 1. One decode step of the deployed cache, schematically. Blue: the GPU's own kernels. Grey: host reads of the misses, which the step waits for. Orange: the background copy that fills an admitted expert's slot, a second read of the same expert.

admits whenever it may; the *fewest-admission* one admits as rarely as any of them. An admitted expert is read *once* (1R), copied into its slot in the step that needs it, or *twice* (2R), served by the CPU in that step and copied in the background. *CPU only* marks a configuration run with the in-step fetches off: every miss the cache does not admit is served by the CPU. A copy issued a few steps before the expert's use is read *ahead*. *Oracles* read the future routing from a trace recorded on the same machine. Configurations are named by the set admitted and the number of reads: MIN-1R is MIN's greedy set read once, MIN-2R the same set read twice, Dep-1R the deployed set read once, and Few-1R and Few-2R MIN's fewest-admission set. Table II lists the configurations, machine sets and yardsticks (the window policies of Section VIII are in Appendix G), and Appendix C maps the codes of the appendix

tables to these names.

Machines: Every machine is a rented host with one RTX 5090 (an RTX 4090 in Section VII). Each runs our bandwidth probe before any model run. It reads host memory from CPU threads, at the number of helper threads the engine uses (B_c), and over PCIe into the GPU, at the rate the engine's copies use (B_p); B_{host} is its highest reading by any method. The *link-to-CPU ratio* is B_p/B_c . We call a link *fast* when the ratio is at least 0.5 and *slow* otherwise; the term compares the link with the CPU, not with other links, and we drew the 0.5 line after the panel's data, before the registered tests of Sections IV, VI and VII. In the last registered test, three idle reruns on each of its two valid machines came within 1.1% of the first run's best reading. Ryzen, Core and Threadripper parts, high-end desktop ones included, are *consumer* processors; EPYC, Xeon and engineering-sample parts are *server* processors.

Statistics and registration: Time per token is averaged over problems, and speed ratios are ratios of mean times. Machines differ far more than problems (two panel machines with the same CPU model differ by up to 29% in the deployed cache's time), so the machine is the unit: claims are orderings that hold on every machine of a stated class, summaries over machines carry 95% *t*-intervals over machines [21], and comparisons on one machine carry 95% paired-bootstrap intervals over problems. A *launch* is one rental, run in two or three processes, *rounds* (one at 25% in the registered tests); a machine *ran stably* when its outputs matched the reference and its deployed cache's time per token agreed within 2% between rounds. A speed ratio moved by a median 0.2% between rounds and 0.5% between rentals, and Appendix D rescores the

registered clauses with both. Every experiment’s predictions were committed to a public branch before its machine started; Appendix D scores them.

III. HOW FAST COULD IT BE?

Consider any system that executes the model’s exact routing with at most C experts of each layer resident on the GPU. Per token, each of the Lk routed experts is served by a resident copy or read from host memory. Let R^* be MIN’s host reads per token, summed over layers, on the routing trace the engine runs, each step’s requests served as one set (Appendix N); no policy, even one that knows the future, reads fewer. If m experts per token run on the CPU, host memory supplies at least $\max(R^*, m)S$ bytes for experts of S bytes, and the GPU reads the dense weights D plus $(Lk - m)S$. With everything overlapped,

$$\bar{T} \geq \min_{0 \leq m \leq Lk} \max\left(\frac{D + (Lk - m)S}{B_{\text{gpu}}}, \frac{\max(R^*, m)S}{B_{\text{host}}}\right). \quad (1)$$

At a host-bound budget this is MIN’s read time, R^*S/B_{host} . The bound assumes exact routing, whole experts, at most C residents per layer and one token per step. It drops all latencies, lets the two read paths share B_{host} , and is relative to the probe: a reader faster than the probe could beat it (Section XI). None of the 302 times per token behind Table IV (25 launches, gpt-oss 11% and 25%) is below it; the fastest, the read-ahead oracle’s, is $1.33\times$ it.

The demand bound: reads that wait for the router: A system that reads an expert only after its layer’s router has chosen it cannot overlap that read with the GPU’s work: layer l ’s misses are known only after its attention and router, and layer $l+1$ starts only after layer l ’s experts. Its time is at least the GPU’s non-expert work plus the reads, in series:

$$\bar{T}_{\text{demand}} \geq T_{\text{GPU}} + \frac{R^*S}{B_{\text{host}}}, \quad (2)$$

where T_{GPU} is the GPU’s attention, dense layers, norms and routers per token: 2.9 ms for gpt-oss, the smallest of the 14 Nsight profiles of our engine we use; a system with faster non-expert kernels has a lower bound. Every system we compare reads on demand; only reads issued before routing can go below Eq. (2).

The ordered bound: reads foresight cannot free: Foresight does not free every read. An expert run on the CPU needs its layer’s attention output, and the next layer’s attention needs its result, so CPU reads sit in series with the GPU’s non-expert work; only copies over the link can be issued ahead. This assumes that the CPU reads an expert’s weights from memory when it runs it; a read-ahead into the CPU’s cache would relax it. With a share f of MIN’s reads served by the CPU at the probe’s highest CPU reading $\hat{B}_c$ and the rest copied at its highest link reading $\hat{B}_p$,

$$\bar{T} \geq \min_{0 \leq f \leq 1} \max\left(\frac{(1-f)R^*S}{\hat{B}_p}, T_{\text{GPU}} + \frac{fR^*S}{\hat{B}_c}\right), \quad (3)$$

and Eq. (1) holds as well, so the bound is the larger of the two. On the median consumer machine at gpt-oss 11%, Eq. (3)

equals Eq. (1) (ratio 1.00); where the link is slow, it is up to 1.56 times as high.

The read time is nearly reachable: A microbenchmark replays MIN’s per-layer reads of gpt-oss on 3 machines without the model: CPU threads read part of each layer’s experts while the copy engine copies the rest. It reaches 86–96% of the bound’s read time (Table S25; registered: at least 50%, a loose threshold). The GPU term of Eq. (1) is looser, but at datasheet rates it binds only at the largest budgets and at gpt-oss 25% on the machines with the fastest host memory (Table S24); at the GPU’s measured batch-1 rate it also binds at gpt-oss 25% on both hosts of Table III and at Qwen3 25% on host B.

Where systems stand: Table III places three systems against the bounds on two machines. Our cache leads Free-Token [9] at 11 of 12 configurations and stock llama.cpp at all of them (Appendix J). Across 25 machines with consumer processors at gpt-oss 11%, it runs at 31–54% of the speed of Eq. (1) (median 47%, quartiles 44–48%), the bound we use throughout (40–54% of the ordered bound, Eq. (3)); the 5 new machines on the deployed path in Section IV fall at 41–48%. With each machine’s rate taken as the larger of its probe’s best reading and the rate its engine’s reads imply (higher on 4 machines, by up to $1.16\times$; Table S23), the range is 31–52%.

Published systems (post hoc): Scored against the same bound on their papers’ own hardware at datasheet rates, 20 published batch-1 measurements reach a median 13.6%; ours, scored the same way, a median 27% (caveats in Appendix M).

IV. WHERE THE SECONDS GO

The gap is the deployed cache’s time beyond Eq. (1). Oracles in the engine remove three parts of it: what is cached (MIN’s set instead of the deployed one), how it is read (each admission read once, copied in the step, instead of served by the CPU and copied in the background), and when (copies issued a few steps early). Because the deployed read path’s in-step fetches put misses the cache did not choose into its slots, we measure the first two changes on two read paths: with the in-step fetches off, on 5 machines rented for two registered tests, and with the fetch table on, as deployed, on the 15 panel machines and 5 machines rented for an earlier registered test (Fig. 2 and Tables IV and V).

With the fetches off, MIN’s set pays alone: At gpt-oss 11%, MIN’s set read twice ran 1.09 – $1.16\times$ the deployed cache on the same path and closed a mean 19% of the gap (interval over machines 13–25%, as registered); the deployed set read once ran 1.02 – $1.07\times$ and closed 8%; MIN-1R, MIN’s set read once, closed 36%. Together the two changes paid more than apart on four of the five machines, as registered, but not on the Core i9-14900K (-0.3 ms per token, interval $[-0.7, 0.1]$; its rounds disagreed, Appendix D). The extra was 21% of the gap on the two Ryzen 9 5950Xs (ratio 0.78–0.79, one provider) and -3 to 3% on the others (0.54–0.57; post hoc). MIN’s set admits 20.4 experts per token against the deployed set’s 8.2 on every machine, so reading once saves it more reads; what a saved read is worth depends on the link. At 25%, MIN’s set

TABLE III

DECODE SPEED AT EQUAL GPU EXPERT MEMORY ON TWO RTX 5090 HOSTS (30 AIME-25 PROBLEMS, THE FIRST 256 GREEDY TOKENS, ONE SESSION).

Model	Experts on GPU	llama.cpp (tok/s)	FreeToken (tok/s)	Ours (tok/s)	Ours $\div$ FreeToken (95% CI)	Ours $\div$ llama.cpp	Bound, datasheet tok/s	GPU ours	Bound, measured tok/s	GPU ours
Host B: Ryzen 9 9950X3D, card memory 17001 MHz; CPU 72, link 53, together 78, highest 88 GB/s (the bound's rate)										
gpt-oss	11%	34.9	54.0 ^h	69.9	1.29 [1.28, 1.31]	2.00 $\times$	172	41%	172	41%
gpt-oss	25%	40.0	85.6 ^h	109.2	1.27 [1.25, 1.29]	2.73 $\times$	430	25%	279	39%
gpt-oss	40%	47.7	132.2 ^o	152.5	1.15 [1.13, 1.17]	3.20 $\times$	519	29%	279	55%
Qwen3	12.5%	19.5	38.7 ^h	40.0	1.03 [1.02, 1.04]	2.05 $\times$	96	42%	96	42%
Qwen3	25%	22.1	54.7 ^h	63.1	1.15 [1.13, 1.17] [†]	2.86 $\times$	214	30%	192	33%
Qwen3	43.75%	28.4	103.1 ^o	108.2	1.05 [1.04, 1.06]	3.81 $\times$	308	35%	192	56%
Host S: Ryzen 9 9950X, stock clock, card 14001 MHz; CPU 56, link 53, together 66, highest 71 GB/s; FreeToken backends as on B										
gpt-oss	11%	28.9	48.0 ^h	57.9	1.21 [1.19, 1.22]	2.00 $\times$	140	41%	140	41%
gpt-oss	25%	33.6	78.7 ^h	94.1	1.20 [1.16, 1.22]	2.80 $\times$	351	27%	275	34%
gpt-oss	40%	40.1	122.2 ^o	133.6	1.09 [1.08, 1.11]	3.34 $\times$	514	26%	275	49%
Qwen3	12.5%	16.1	32.7 ^h	33.5	1.03 [1.02, 1.04]	2.08 $\times$	78	43%	78	43%
Qwen3	25%	18.5	51.4 ^h	53.9	1.05 [1.04, 1.06]	2.91 $\times$	174	31%	174	31%
Qwen3	43.75%	23.8	98.3 ^o	95.8	0.97 [0.96, 0.99]	4.02 $\times$	305	31%	189	51%

Ratios are of mean speeds; the ratio over FreeToken is paired by problem within one launch, with a 95% bootstrap interval over problems. Host B's rows come from three rentals of the same machine (the same GPU), so its ratios over llama.cpp compare launches. Our cache and FreeToken ran twice per cell and the second launch is shown (on host B both launches ran ours first); llama.cpp and the [†] cell ran once. Each system decodes its own greedy text. Our cache uses each machine's fetch table; FreeToken uses its faster backend per budget (^h hybrid, ^o offload), picked on host B on a separate launch. gpt-oss: gpt-oss-120b; Qwen3: Qwen3-30B-A3B. *Bound*: Eq. (1) on that machine (MIN's reads, the host's highest probed rate) with the GPU at its datasheet rate, and at the rate all-in-VRAM decode measures at batch size 1 (52% of datasheet for gpt-oss, 61% for Qwen3; Table S24 varies the other assumptions; Appendix J gives the pooled bounds and all-in-VRAM speeds). [†] Selection launch only, which favours FreeToken.

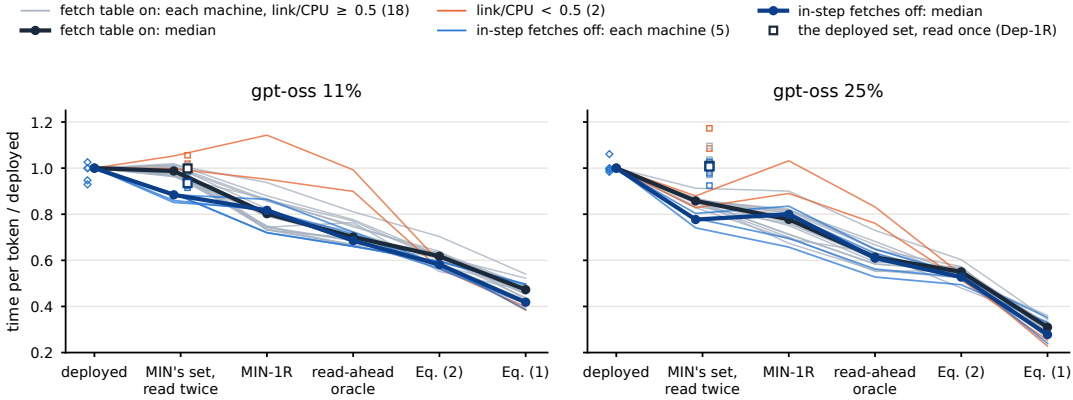


Fig. 2. Where the seconds go, measured: each state's time per token relative to the deployed cache on the same machine (lower is faster) as what is cached changes (MIN's set, read twice; squares: the deployed set, read once), then both, then when, with the two bounds. One line per machine. Grey (orange: link-to-CPU ratio below 0.5): the deployed read path, where the second state is MIN's schedule with the fetch table, not its set held. Blue: in-step fetches off (diamonds: the deployed cache without them). Thick: medians over fast-link machines.

read twice closed 33% and the extra was about zero (-0.1 to 0.1 ms).

The in-step fetches displace the set the cache chose:

On the four of these machines whose probe gave a nonzero fetch table, MIN-2R missed 48.1–50.8 experts per token at 11% with the in-step fetches and 43.5 without them: the table copies experts that MIN bypasses into slots that MIN keeps. Most of what Dep-1R copied in the step was the table's too: 20.8–32.0 experts per token, of which its own admissions were 8.2. MIN-1R's copies follow MIN's schedule, which replaces the table, so MIN-1R holds MIN's set on either path.

With the fetch table on, MIN's set pays only when read once: On the deployed path at 11%, MIN-2R closes 0% of

the gap, Dep-1R 0%, and MIN-1R 33% on the panel (39% on its 13 fast-link machines) and 35% on the new machines. We registered that this interaction would be positive on every machine; it was, on 9 of the 10 machines of the panel's main job and on all 5 new ones. On the machines with a nonzero table, turning the in-step fetches off shrank the interaction's share of the gap from 19–52% to -3 to 21%; on the one whose table was all zero (a Ryzen 9 5950X), it was 22% with the table and 21% without. Most of the interaction on the deployed path is therefore the table's.

What is left: The read-ahead oracle closes another 15% at 11%, and about half the gap remains (Table IV). On fast links two named parts account for it (at 25% they overshoot slightly).

TABLE IV

THE DEPLOYED CACHE’S GAP TO EQ. (1), SPLIT BY ORACLES IN THE ENGINE (FIG. 2): MEAN SHARE OF THE GAP, IN PERCENT, WITH 95% t -INTERVALS OVER MACHINES.

Part of the gap	gpt-oss 11%			gpt-oss 25%		
	Fetch table on			Fetch table on		
	Fetches off (5)	Panel (15)	New (5)	Fetches off (5)	Panel (15)	New (4)
<i>Measured</i>						
What is cached: MIN’s set, read twice (MIN-2R)*	19 [13, 25]	0 [−2, 3]	3 [−1, 8]	33 [24, 42]	21 [19, 22]	20 [18, 21]
How it is read: the deployed set, read once (Dep-1R)	8 [2, 14]	0 [−2, 2]	0 [−2, 1]	3 [−8, 13]	−3 [−7, 1]	−2 [−3, −1]
Both: MIN’s set, read once (MIN-1R)	36 [13, 58]	33 [21, 44]	35 [22, 48]	36 [15, 56]	31 [23, 39]	27 [23, 32]
Then the read-ahead oracle	18 [10, 25]	15 [11, 19]	19 [14, 24]	23 [19, 27]	21 [18, 23]	24 [21, 27]
Left after all three	43 [31, 56]	52 [43, 62]	46 [38, 55]	42 [29, 55]	48 [41, 55]	48 [44, 53]
The in-step fetches themselves†	3 [−5, 12]	−	−	−1 [−7, 5]	−	−
<i>Attributed: of what is left</i>						
the GPU’s non-expert time (T_{GPU})	25 [18, 31]	27 [25, 30]	28 [20, 36]	32 [24, 40]	34 [31, 37]	38 [33, 43]
the oracle’s reads beyond MIN’s, at B_{host}	18 [14, 23]	21 [20, 23]	18 [15, 20]	18 [13, 23]	19 [18, 21]	16 [15, 17]
residual	0 [−11, 12]	4 [−6, 13]	1 [−3, 4]	−8 [−18, 2]	−5 [−12, 2]	−5 [−7, −3]

Panel: the 15 machines that ran every state before the registered tests (exploratory). *New*: the 5 machines rented for the registered test (population and predictions fixed before any started; the deadline cut one 25% round). *Fetches off*: the machines of the registered tests of Table V, whose arms ran with the in-step fetches off; their shares are of each machine’s gap with its fetch table, like the other columns. *On the deployed path (Panel, New) the in-step fetches displace MIN’s set, so this row measures MIN’s schedule with the fetch table, not the set held; in the fetches-off column it measures the set held. †The deployed cache with its fetch table against without it. *Both, then the read-ahead oracle* and *left* sum to the gap (in the fetches-off column, with †). The last three rows split *left* by attribution: T_{GPU} (2.9 ms) and the read-ahead oracle’s reads beyond R^* at B_{host} ; the residual is what they leave (Appendix L gives the probe sensitivity).

TABLE V

THE 2×2 WITH THE FETCH TABLE ON AND WITH THE IN-STEP FETCHES OFF, REGISTERED (GPT-OSS 11%).

Machine	Ratio	ms per token		Deployed off/on	MIN-2R		Dep-1R		MIN-1R	Interaction (ms)
		Dep.	Eq. (1)		on	off	on	off		
Ryzen 9 5950X [‡]	0.79	29.7	14.7	1.00±.00	1.12±.01	1.12±.01	1.06±.01	1.07±.00	1.39±.02	3.2 [3.0, 3.3]
Core Ultra 7 265K	0.57	19.3	8.1	1.06±.01	1.00±.01	1.10±.01	1.01±.00	1.03±.00	1.16±.01	0.4 [0.3, 0.5]
Ryzen 9 5950X	0.78	29.7	14.8	1.00±.01	1.01±.01	1.12±.01	1.01±.00	1.07±.01	1.39±.01	3.1 [2.9, 3.3]
Core i9-14900K [§]	0.54	15.1	5.9	0.98±.02	1.03±.01	1.16±.03	0.99±.00	1.04±.02	1.19±.02	−0.3 [−0.7, 0.1]
Core Ultra 9 285K [§]	0.54	19.4	8.0	1.08±.01	0.99±.01	1.09±.01	1.00±.00	1.02±.00	1.13±.01	0.3 [0.2, 0.3]

Speeds relative to the deployed cache on the same machine and read path (*on*: the deployed read path, with the fetch table; *off*: the in-step fetches off). MIN-1R’s forced plans replace the table, so it is the same on both paths; its column is against the deployed cache with the fetches off. *Dep.*: the deployed cache with its fetch table (*on*); Eq. (1): the bound. Subscripts: half-widths of 95% intervals over problems. Interaction: MIN-2R’s plus Dep-1R’s time minus the deployed cache’s and MIN-1R’s, fetches off, in ms per token, with its 95% interval. [‡]An all-zero fetch table: no in-step fetches on either path. [§]The second test.

The oracle still reads its remaining misses after their router runs, so the GPU’s non-expert time stays in series with them; and it reads more than MIN, because its early copies evict sooner. Removing both would take every read issued before routing at MIN’s count, and only copies over the link can be issued early (Eq. (3)). On slow links the two parts fall short: at 11% a residual of 35–47% of the gap stays unexplained on the two slow-link panel machines, and 31–36% on the three slow-link RTX 4090s of Section VII; measuring against Eq. (3) does not remove it (Appendix L).

V. HOW FORESIGHT MUST BE SPENT

This section measures each way of spending foresight as a speed relative to the deployed cache on the same machine. A running example: on a Ryzen 9 9950X at gpt-oss 11%, Eq. (1) gives 10.0 ms per token and Eq. (2) 13.0. The deployed cache takes 20.6 ms, MIN-1R 15.2, and the read-ahead oracle 13.8.

Reading each admitted expert once pays most at the smallest budget: MIN-1R is faster than the deployed cache at

every budget on the three factorial machines (Fig. 3). The usual ways of spending foresight waste reads. Prefetching without bypass, Belady’s original rule, admits experts MIN would bypass. Serving a miss on the CPU before copying it reads every admission twice. At the smallest budget of each model these gain at most 15% or lose, a trade-off known for disks [22] and CPU caches [23]. These gains were registered as bands per budget on two machines and held at most budgets. With the in-step fetches off, on the five machines of Section IV, MIN’s set gains read twice as well; read once, it ran 1.02–1.24× that at 11%, but at 25% (one round) gained only on the two Ryzen 9 5950Xs (ratio 0.78–0.79; 1.12–1.13×) and lost on the three Intel machines (0.54–0.57; 0.96–0.97×; post hoc).

Admit as rarely as optimality allows: The greedy schedule admits whenever MIN allows and may evict an expert again before its next use. The fewest-admission schedule makes 0.60–0.73 of the greedy one’s copies for the same hits in replay. Read once in the step at gpt-oss 11% (Few-1R), it beat the deployed

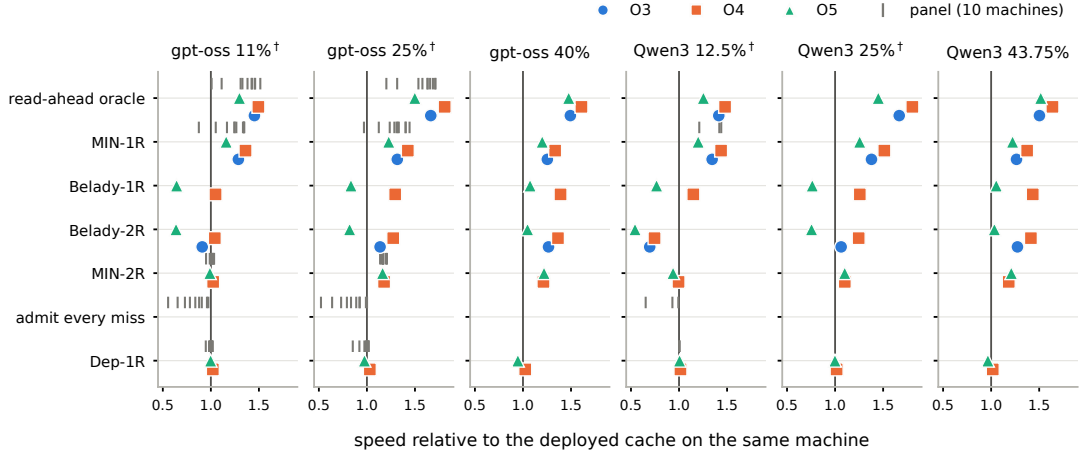


Fig. 3. What foresight is worth, by how it is spent: speed of each configuration of Table II relative to the deployed cache on the same machine, at the six budgets ([†]host-bound). Three machines at every budget (O3–O5: two Ryzen 9 9950X and a 9950X3D); the 10 machines of the panel’s main job (grey ticks) at the two host-bound gpt-oss budgets, and 3 of them at Qwen3 12.5%. The 95% intervals are mostly narrower than the markers. MIN-1R gains at every budget on these machines; reading twice or admitting what MIN bypasses gains at some budgets and loses at others.

cache on all 15 machines that ran it stably, by a geometric mean of $1.26\times$ (registered on 7 of them; two are RTX 4090s and two have server processors). Of the four machines that ran it unsteadily or failed a validity check, it lost on two, the two with the slowest links (ratios 0.14 and 0.21; Appendix I), and gained on the others.

VI. WITHOUT FORESIGHT, THE ENGINE RECOVERS LITTLE

The engine has two mechanisms that act on the same levers without foresight. A larger admission margin admits less: it cuts the deployed cache’s host reads by 4–6% at 11%. A layer-ahead copy applies the next layer’s router to this layer’s input and copies the top predicted expert ahead, so that a correctly predicted admission is read once. Four variants combine the two (Table II). We registered that none would run faster than $1.10\times$ the deployed cache, or capture more than 35% of the read-ahead oracle’s gain, on any machine.

At 11% the best variant runs $1.007\text{--}1.024\times$ the deployed cache, at most 6% of the read-ahead oracle’s gain. The counters say why: the variants read about as many host experts per token as the deployed cache ($1.57\text{--}1.76 R^*$). A prediction one layer ahead moves reads earlier but does not avoid them. On the slow-link RTX 4090s of Section VII, the variants with the layer-ahead copy lose time ($0.64\text{--}0.72\times$ at 11%), as registered, and the larger margin alone is the best (up to 26% of a smaller oracle gain).

VII. THE LINK-TO-CPU RATIO ON A SECOND CARD

On the RTX 5090 machines, the gain of every configuration that moves reads from the CPU to the link rises with the link-to-CPU ratio (Fig. 4; for MIN-1R, Spearman 0.89 over 19 machines). We froze that trend on the panel, one line in the log of the ratio per configuration, and registered that it would predict the oracles’ speed on RTX 4090 machines, a slower card behind a PCIe 4.0 link, within 0.10 in log.

It did, on all five RTX 4090 machines of two tests, at ratios 0.37–0.77: MIN-1R and the read-ahead oracle came within 7% of the prediction at 11% and 5% at 25% (Table VI); every machine’s 95% interval lies inside the band, and the interaction of Section IV is positive on the two with fast links, as registered. Predicting no change would have missed by up to 46% at 11% and 69% at 25%. The trend is coarser on fast links: applied after the fact to the new RTX 5090s of Section IV, it misses five of their 18 cells by more than the band, by up to 18%. The ratio sets the direction of the gain more reliably than its size. On the two machines whose link reads at under a third of the CPU rate (0.28–0.32), Few-2R paid instead of MIN-1R ($1.11\text{--}1.22\times$).

VIII. HOW MUCH FORESIGHT IS NEEDED

The oracles know the whole future; a predictor would know part of it, imperfectly. We give the cache a *window*, the routing of the next W tokens, exact or with each future expert right only with probability r . Throughout this section, the value of a window is its share of MIN’s gain over admitting every miss, the online policy that reads least at these budgets (Appendix G gives the policies).

Next-token prediction buys little: Exact routing of the next token recovers at most 0.31 of MIN’s gain at any host-bound budget of our two models; half of the gain at gpt-oss 11% needs 4 tokens (Fig. 5). In the engine, an exact 16-token window recovers 0.78–0.93 of the gain in time at 11% (registered at 0.80 or more; the lowest machine missed). Against the deployed cache, the same window copied in the step runs $0.89\text{--}1.32\times$ at 11% on 4 machines, below 1 on the 2 with the lowest link-to-CPU ratios. Accuracy matters as much as horizon: an 8-token window that is right half the time recovers 0.29 of the gain at 11%, about what 2 exact tokens buy (0.31).

The horizon is measured in distinct experts (exploratory): On 9 models’ own sampled text, the window that closes half of the read gap ranges from 0.6 to 34 tokens. Paging theory

TABLE VI

THE REGISTERED TESTS ON MACHINES RENTED FOR THEM: SPEED RELATIVE TO THE DEPLOYED CACHE ON THE SAME MACHINE (GEOMETRIC MEAN OF ITS ROUNDS) OF MIN-1R, THE READ-AHEAD ORACLE (*ahead*), AND THE BEST OF THE FOUR VARIANTS WITHOUT FORESIGHT (*none*: MARGIN, LA, LA-1R, LA-1R-MARGIN; SECTION VI), AND THAT VARIANT’S CAPTURE, ITS SHARE OF THE READ-AHEAD ORACLE’S GAIN.

Machine	Ratio	gpt-oss 11%				gpt-oss 25%			
		MIN-1R	Ahead	None	Capture	MIN-1R	Ahead	None	Capture
RTX 5090, rented for the test of policies without foresight									
Core Ultra 9 285K	0.54	1.22±.01	1.46±.01	1.02±.00	4±0%	1.24±.01	1.64±.02	1.03±.00	4±0%
Core i9-13900KF	0.59	1.25±.01	1.46±.01	1.01±.00	2±0%	1.26±.01	1.61±.02	1.01±.01	2±1%
Ryzen 9 5950X	0.84	1.37±.01	1.50±.01	1.01±.00	2±0%	—	—	—	—
Ryzen 7 9800X3D	0.88	1.15±.01	1.34±.01	1.01±.01	4±2%	1.22±.01	1.55±.02	1.01±.00	2±1%
Ryzen 9 7945HX	0.89	1.24±.01	1.42±.01	1.02±.01	6±1%	1.28±.01	1.63±.02	0.99±.01	−1±1%
RTX 4090, the second-card test									
Core i9-14900KF	0.37	0.96±.00 [0.99]	1.13±.01 [1.11]	1.03±.00	26±3%	1.02±.00 [1.07]	1.30±.01 [1.29]	1.04±.00	13±1%
Core Ultra 9 285K	0.39	1.01±.00 [1.01]	1.20±.01 [1.13]	1.04±.00	20±2%	1.06±.01 [1.09]	1.36±.01 [1.31]	1.04±.00	12±1%
Core Ultra 9 285K	0.40	1.01±.00 [1.01]	1.20±.01 [1.13]	1.04±.00	20±2%	1.06±.01 [1.09]	1.37±.01 [1.32]	1.05±.00	12±1%
RTX 4090, the second-card test at higher ratios									
Ryzen 7 7800X3D	0.57	1.13±.01 [1.14]	1.30±.01 [1.26]	1.01±.01	4±2%	1.17±.01 [1.21]	1.48±.01 [1.47]	1.02±.00	5±1%
Core i5-12400	0.77	1.31±.01 [1.25]	1.46±.01 [1.37]	1.01±.00	3±1%	1.35±.01 [1.31]	1.69±.02 [1.61]	1.02±.00	3±0%

Ratio: link-to-CPU. RTX 5090s: their population (desktop-class, link-to-CPU ratio at least 0.5) was fixed before any of them started. RTX 4090s: the predictions were committed before any RTX 4090 ran, and the same predictions were tested again on machines at higher ratios; in brackets, what the RTX 5090 machines’ trend in the link-to-CPU ratio predicted for that machine. Subscripts: half-width of the 95% interval over problems (paired bootstrap within the machine).

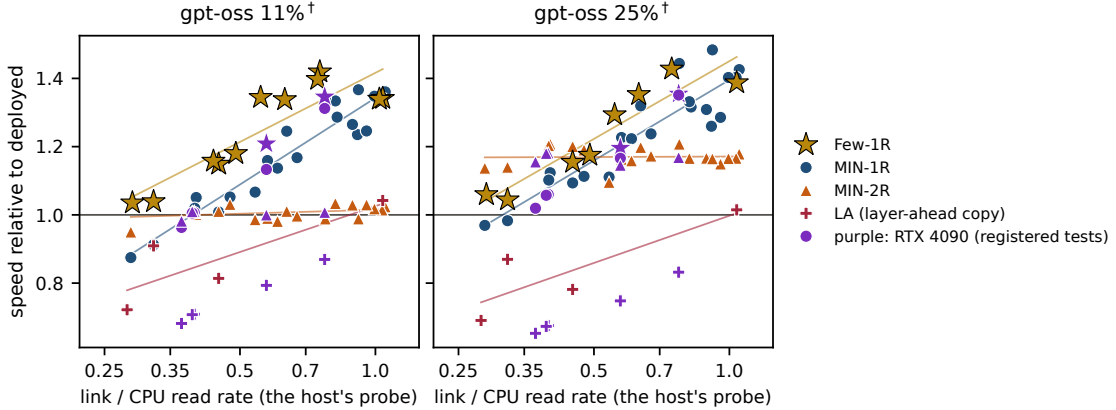


Fig. 4. The machine decides which way of spending foresight pays: speed relative to the deployed cache on the same machine against the machine’s link-to-CPU ratio, at the two host-bound gpt-oss budgets. Every marker is a machine that passed our checks. Stars: Few-1R. Circles and triangles: MIN-1R and MIN-2R, both on the greedy schedule. Crosses: the deployed cache with the layer-ahead copy. Purple markers: the RTX 4090 machines of the registered second-card test (Section VII), not used in the fits. Lines: least-squares fits in the log of the ratio, drawn for the trend only.

measures lookahead in distinct pages [24], [25], and so does the rule that fits: within that window a layer requests about $0.65C$ distinct experts (0.66 – $0.81C$ on the AIME routing the engine runs), and it predicts a model’s horizon as well as a two-constant power law in tokens (Appendix F).

Nothing realisable reaches it: Forecasters trained on routing histories, linear or recurrent, close at most 15% of the read gap on gpt-oss: their precision falls with the horizon. Speculative batches save an online cache about a tenth of its reads at 90% draft acceptance and cost reads at 80% [26]–[28] (Appendix F). Layer-ahead predictors [11], [12] reach 84% recall in our engine but see less than one token ahead (Section VI).

IX. WHAT THIS MEANS FOR SYSTEM BUILDERS

These rules come from one engine (ours), batch-1 decode and one prompt set, and rules 2–4 from one model at two budgets; elsewhere they are hypotheses to test, with Eq. (1) as the yardstick.

- 1) **Report the distance to the machine’s bound.** A speed-up over a chosen baseline hides how much is left; Eq. (1) needs only a routing trace and a probe.
- 2) **Make the read path keep the set the policy chose.** In-step fetches that put unchosen misses into slots undo a better set: with them, MIN’s set read twice closes none of the gap at the smallest budget; without them, 13–23% (five machines, gpt-oss 11%).
- 3) **Do not expect a policy without foresight to get there.** Admitting less and predicting one layer ahead recover

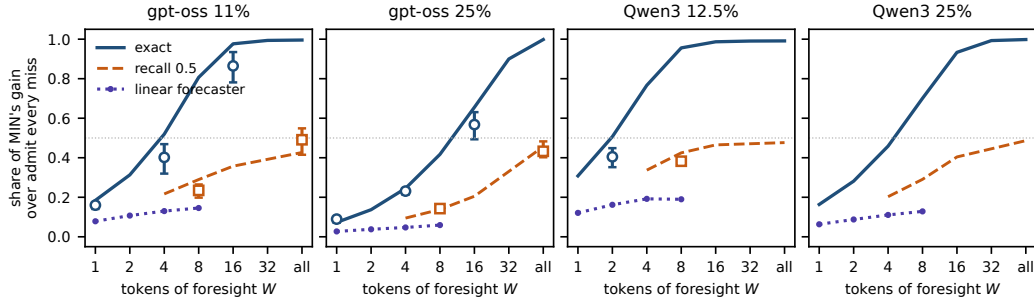


Fig. 5. Baseline here: *admit every miss*, not the deployed cache. The share of MIN’s gain over admitting every miss recovered, in host reads on the routing engine runs, by exact routing of the next W tokens (solid), by a window whose experts are each right with probability 0.5 (dashed) and by a linear forecaster fitted on other text (dotted). Markers: the same share in time per token (mean and range over the panel’s main-job machines: 10 for gpt-oss, 3 for Qwen3 12.5%). Grey line: half of the gain.

at most 6% of foresight’s worth where it is worth most.

- 4) **Choose the read path by the link-to-CPU ratio and the budget.** Where the link is fast, copy admitted misses into their slots in the step (at ratios of 0.54–0.57, only at the smallest budget; post hoc, Section V); where it is slow, serve them on the CPU and copy in the background (Section VII).
- 5) **Issue reads before routing, and predict far enough ahead to save reads.** A read that waits for the router stays in series with the GPU’s non-expert time; only a prediction lets it start earlier, and only over the link (Eq. (3)). Predict the next $\approx 0.65C$ distinct experts of each layer, not the next token.

X. RELATED WORK

Offloading systems: Expert caches copy misses over PCIe [3]–[5], [29]; CPU-execution systems run non-resident experts on the host [6]–[8], [30]; FreeToken [9] and DALI [10] split each layer’s misses between the two. Predictors guess upcoming experts from the previous layer [11], [12], several tokens ahead [13], [31], from draft tokens [26]–[28], or decouple the router [32]. Each reports speed-ups over its own baseline; none bounds the machine (more in Appendix K).

Bounds and caching theory: WiSP [33], Budgeting Bytes [34], MoE-CAP [35] and the roofline [36] bound decode by bytes over bandwidth; Zhang [37] and Liang et al. [38] compare caches with MIN in hits. We add the time domain. MIN with bypass maximises hits [19], [20], not speed [18]; our fewest-admission schedule extends offline caching over reuse intervals [39].

XI. LIMITATIONS

- **Hardware, engine, sample and workload.** One engine, ours; RTX 5090s (RTX 4090s in two tests), rented: a convenience sample of cheap offers with few slow links; unlocked clocks; batch-1 decode of two models on AIME prompts also used in development (teacher-forced in the engine); three tests deviated from their registration (Appendix D).

- **The bounds.** Exact routing; per-layer slots (pooling them lowers MIN’s reads by 4.5–18.6%); no latencies; our probe (Section III), whose link-to-CPU ratio moved up to 12% between two rentals of one machine; our engine’s T_{GPU} (RTX 5090).
- **The decomposition and the oracles.** Two budgets of one model; what the oracles leave is attributed, not measured; oracles read a recorded trace; without the in-step fetches, five machines of two kinds.
- **Outputs.** Parity is teacher-forced KL divergence from an all-VRAM reference (mean 0.0019 and 0.0005 nats), not task accuracy; more in Appendix L.

XII. CONCLUSION

MIN’s host reads at a machine’s probed rate bound batch-1 MoE decode from host memory; our cache reaches 31–54% of it. MIN’s set pays where the read path keeps it; policies without foresight recover little.

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